# Package architecture

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**Package Architecture:**

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├— Frontend Team (HTML, CSS, JS) → Folder-1
├— Backend Team (Python, etc.) → Folder-2
└— Database Team (SQL, etc.) → Folder-3

**Definition**
**Package Architecture** is a concept of dividing a large project into multiple smaller, organized **packages**.
Each package may contain related modules, which together make the project easy to manage, test, and maintain.

**Package**

- A **package** is a **folder** that contains one or more **Python files (modules)**.
- It is used to organize related modules together.
- Every package contains a special file called **__init__.py** (it can be empty).
  This file tells Python that the folder should be treated as a package.

Syntax: pip install package_name
**Types of Packages:**

1. **User-Defined Packages** – Created by the user for their project.
2. **Inbuilt Packages** – Already available with Python (e.g., math, random, os, re).

**<u>Module:</u>**

- A **module** is a **Python file** (.py) that can contain **functions, variables, classes, or objects**.
- Modules allow you to reuse code without rewriting it.

**Example:**

import modulename
**Common Inbuilt Modules:**

- math

- random
- calendar
- keyword
- time

**Types of Modules:**

1. **User-Defined Modules**
2. **Inbuilt Modules**

Difference Between Package and Module

| **Feature** | **Package** | **Module** |
|---|---|---|
| Definition | A folder containing multiple modules or sub-packages | A single Python file |
| Structure | Folder (with __init__.py) | File (.py) |
| Installation | Installed using pip install | Imported using import |
| Example | numpy, pandas | math.py, random.py |

**Importing Modules**

You can import modules in three ways:

1. **Using-> import modulename**

```
import math
print(math.sqrt(64))
print(math.factorial(5))
print(math.lcm(2, 8, 16))
print(math.pi)
```

2. **Using-> from modulename import ***

```
from math import *
print(sqrt(64))
print(factorial(5))
print(lcm(2, 8, 16))
print(pi)
```

3. **Using ->from modulename import specific_function**

```
from math import sqrt, factorial
print(sqrt(64))
print(factorial(5))
```

**Useful Functions:**

```
print(len(dir(math)))   # Count of all functions in the module
```

```
print(dir(math))       # Lists all functions and variables in the module
```

**Random Module:**

The random module is used to generate random numbers or choose random elements.

**Import Methods:(Syntax)**

```
import random
from random import *
from random import choice, shuffle
```

**1. randint()**

Generates a random integer between two given numbers.

**Syntax:**

```
import random
print(random.randint(1, 100))
```

**2. random()**

Generates a random float value between **0.0 and 1.0**.

**Example:**

```
import random
print(random.random())
```

**3. choice()**

Selects a random element from a collection (list, tuple, set, string).

Note: Not applicable for dictionaries.

**Examples:**

```
import random
print(random.choice('python'))
print(random.choice([10, 20, 30]))
print(random.choice((10, 20, 30, 40)))
print(random.choice({10, 20, 30, 40}))
```

**4. shuffle()**

Shuffles the elements of a mutable sequence (like a list).

Note: Cannot be used with immutable data types (tuple, string, set).

**Example:**

```
import random
a = [10, 20, 30, 40, 50]
```

```
random.shuffle(a)
print(a)
```

**User-Defined Modules Example:**

***Create a folder(package)-->create FILE1.py**
**FILE1.py**

```
def valid_un(s):
    if '_' in s:
        return 'Valid username'
    else:
        return 'Not valid username'
def valid_pw(s):
    if len(s) >= 8:
        upper, lower, digit, special = 0, 0, 0, 0
        for i in s:
            if 'A' <= i <= 'Z':
                upper += 1
            elif 'a' <= i <= 'z':
                lower += 1
            elif '0' <= i <= '9':
                digit += 1
            elif i in '@$#':
                special += 1
        if upper >= 2 and lower >= 2 and digit >= 2 and special >= 2:
            return 'Valid password'
        else:
            return 'Not valid password'
    else:
        return 'Password length must be ≥ 8'
```

**FILE2.py** (in the same package)

```
import FILE1
print(FILE1.valid_un('06ita_ishi03'))
print(FILE1.valid_pw('dgHIW$#123'))
```

OR

```
from FILE1 import *
print(valid_un('06ita_ishi03'))
```

```
print(valid_pw('dgHIW$#123'))
```

**Accessing Module from Outside the Package**

```
from Package7AM import FILE1
print(FILE1.valid_un('06ita_ishi03'))
print(FILE1.valid_pw('dgHIW$#123'))
```

**In Short:**

- **Module:** Single Python file (.py)
- **Package:** Folder with multiple modules
- **Import:** Used to bring functionalities
- **From:** Used to access specific parts of a module